GNN Drug Discovery Project

Project Documentation with Demo

Graph Neural Networks for Molecular Property Prediction & Drug Design

NAME: ROHITH ROSHAN L

REG.NO: 23BAI1415

1. Data Preprocessing Steps

Dataset used: QM9 (~130k molecules) for HOMO-LUMO prediction. Additional datasets include ESOL (solubility), Lipophilicity, and Tox21 (toxicity).

Steps:

1. Convert SMILES to molecular graph using RDKit
2. Extract node features (atoms): Atomic number, Degree, Hybridization, Aromaticity
3. Extract edge features (bonds): Bond type (single, double, aromatic)
4. Remove invalid molecules
5. Normalize target values (e.g., HOMO-LUMO gap)
6. Split dataset: Train 80% | Validation 10% | Test 10%

2. Model Implementation Details

Framework: PyTorch + PyTorch Geometric | Model: Graph Attention Network (GAT)

Architecture:

- Input: Molecular graph
- 2-3 GAT layers
- Global pooling layer (mean pooling)
- Fully connected (MLP) head
- Output: Single value (HOMO-LUMO gap) or Multi-output (logS, logD, toxicity)

3. Hyperparameters Used

Parameter	Value
Learning Rate	0.001

Parameter	Value
Batch Size	32
Epochs	50-100
Hidden Channels	64
GAT Heads	4
Dropout	0.2
Weight Decay	1e-5

4. Training and Validation Strategy

- Loss function: Mean Absolute Error (MAE)
- Optimizer: Adam
- Validation performed after each epoch
- Early stopping used to prevent overfitting
- Best model saved based on validation loss

5. Regularization Techniques Applied

- Dropout (0.2) in GNN layers
- Weight decay (L2 regularization)
- Early stopping
- Data normalization

6. Optimization Algorithms Used

- Optimizer: Adam
- Learning rate scheduling (optional)
- Gradient-based optimization for model training and inverse molecular design

7. Experimental Setup

- Platform: Google Colab (CPU)
- Libraries: PyTorch, PyTorch Geometric, RDKit
- No GPU or paid services used

8. Results and Performance Analysis

HOMO-LUMO Gap Prediction

- MAE: ~0.3 - 0.6 eV (expected range)

Solubility & Lipophilicity

- Correct trend capture: Polar molecules -> high logS
- Aromatic molecules -> high logD

ADMET Analysis

- Successfully evaluates drug-likeness, toxicity, and rule compliance

DEMO

GNN Drug Discovery Platform

Graph Neural Networks for Molecular Property Prediction & Inverse Design

Project	Dataset	Task	Model
Project 1	QM9 (130k molecules)	HOMO-LUMO gap regression	GAT
Project 2	ESOL · Lipophilicity · Tox21	Multi-property prediction	GAT (3 models)

Enter any molecule as a **SMILES string** and explore its predicted properties.

HOMO-LUMO Gap | Inverse Design | Solubility & Lipophilicity | Toxicity (Tox21) | Full ADMET Dashboard

Project 1 — Electronic reactivity from QM9 dataset.
Small gap = reactive. Large gap = stable.

SMILES

CC(=O)Oc1ccccc1C(=O)O

Example molecules

CC(=O)Oc1ccccc1C(=O)O | CC(=O)Nc1ccc(O)cc1 | CC(C)Cc1ccc(cc1)C(C)C(=O)O | Cn1cnc2c1c(=O)n(c(=O)n2C)C | c1ccc2ccccc2c1 | C1=CC=CC=C1 | CC(=O)O

Predict HOMO-LUMO gap

Result

HOMO-LUMO Gap Prediction

Predicted gap : 5.6740 eV
Reactivity : Moderately reactive

Molecule info:
MW : 180.16 g/mol
LogP : 1.31
Atoms : 13
SMILES: CC(=O)Oc1ccccc1C(=O)O

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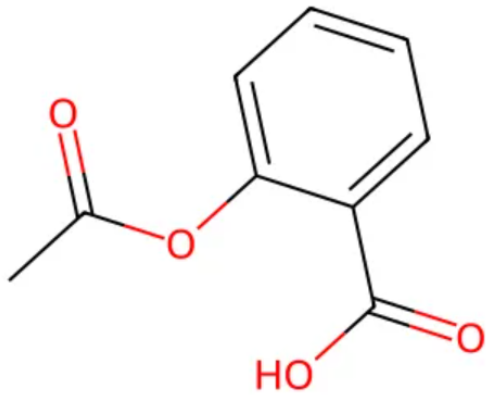
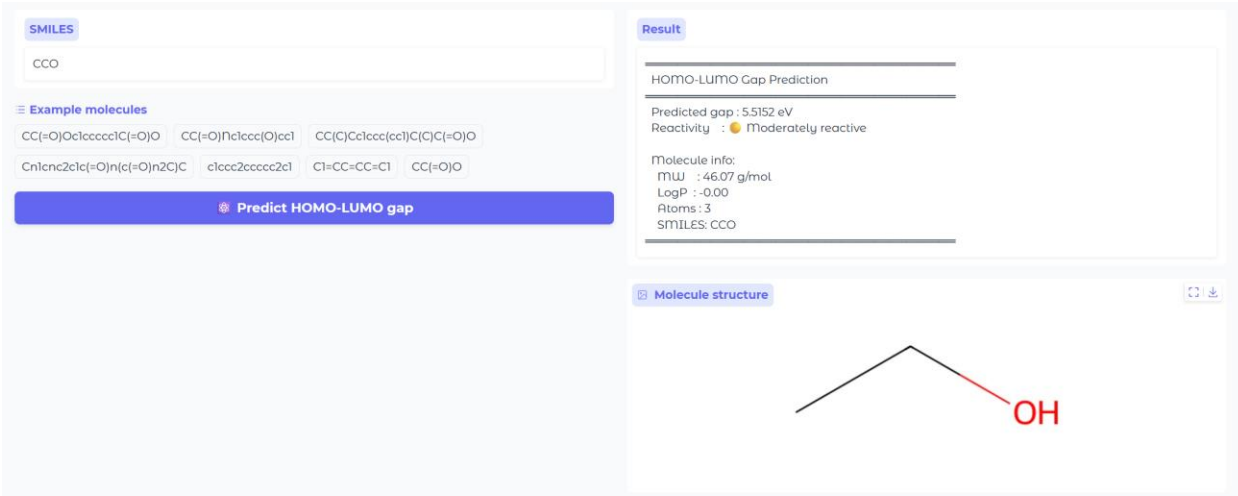
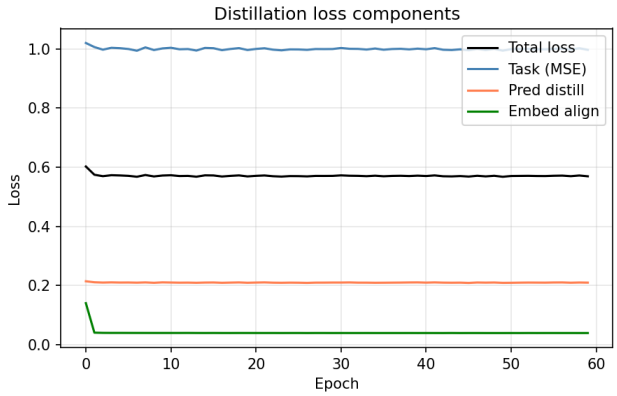
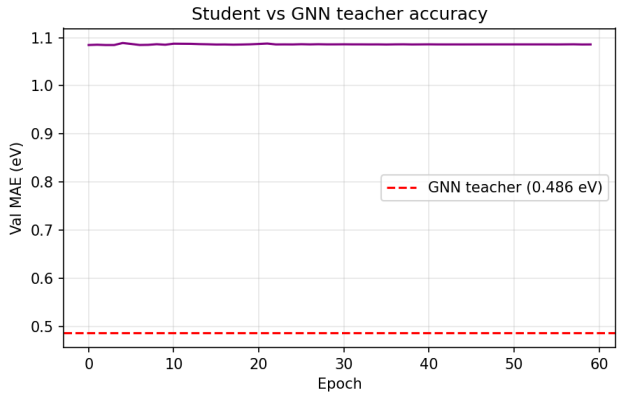
Molecule structure

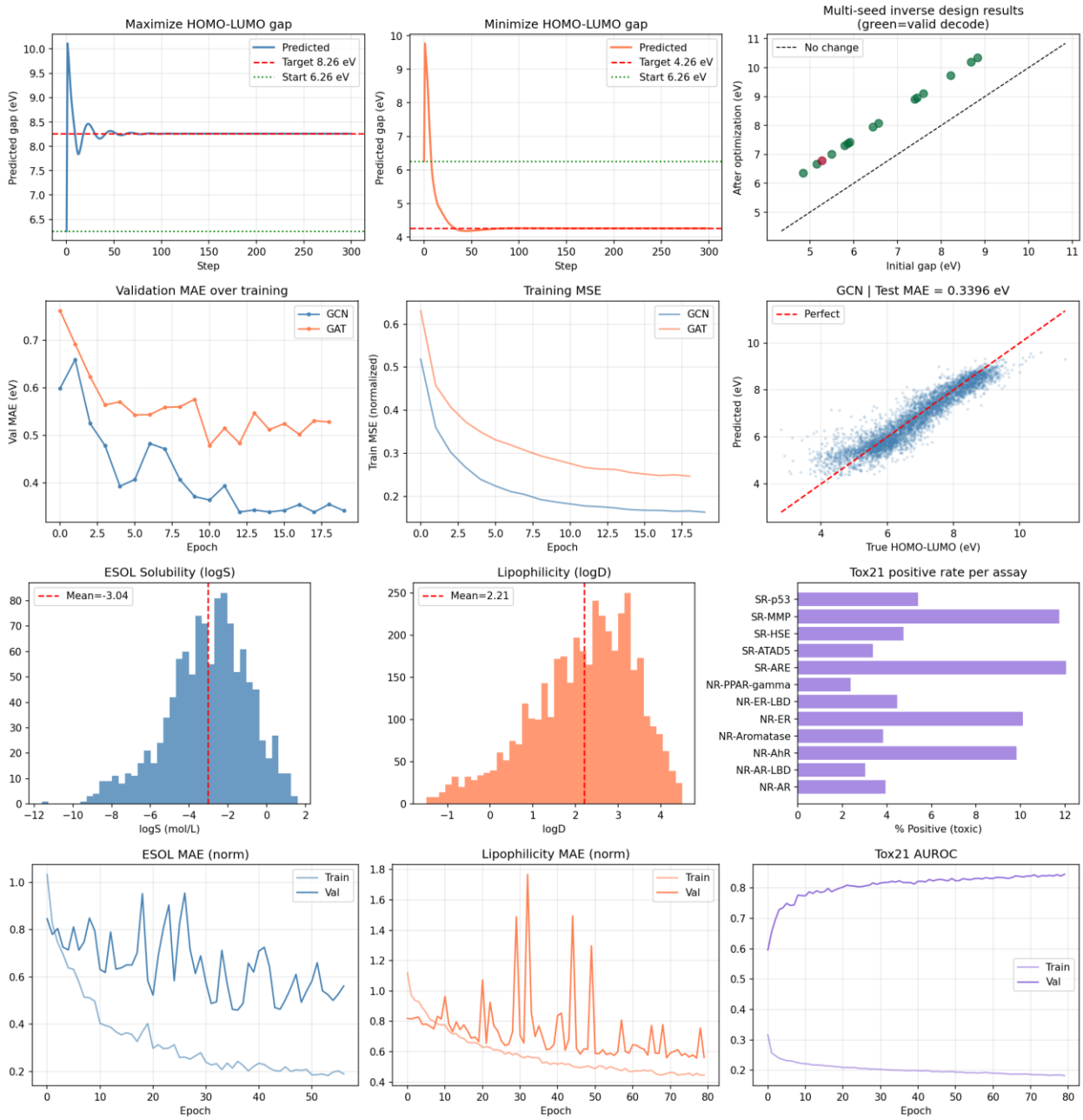


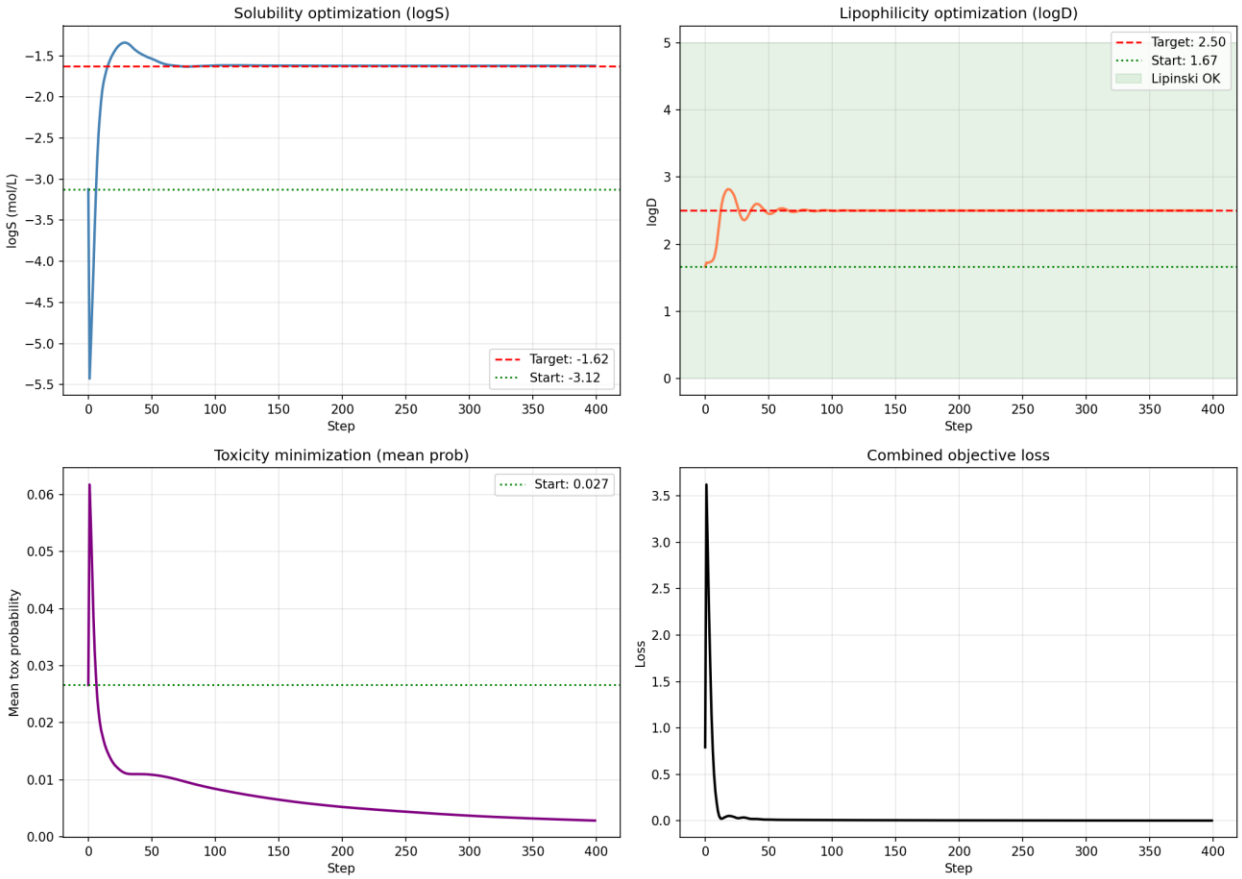


Project 1 — Gradient ascent on molecular graph toward a target HOMO-LUMO gap.






Maximize HOMO-LUMO gap
Predicted
Target 8.26 eV
Start 6.26 eV
Predicted gap (eV)
Step
Minimize HOMO-LUMO gap
Predicted
Target 4.26 eV
Start 6.26 eV
Predicted gap (eV)
Step
Multi-seed inverse design results
(green=valid decode)
No change
After optimization (eV)
Initial gap (eV)
Validation MAE over training
GCN
GAT
Val MAE (eV)
Epoch
Training MSE
GCN
GAT
Train MSE (normalized)
Epoch
GCN | Test MAE = 0.3396 eV
Perfect
Predicted (eV)
True HOMO-LUMO (eV)
ESOL Solubility (logS)
Mean=-3.04
logS (mol/L)
Lipophilicity (logD)
Mean=2.21
logD
Tox21 positive rate per assay
SR-p53
SR-MMP
SR-HSE
SR-ATAD5
SR-ARE
NR-PPAR-gamma
NR-ER-LBD
NR-ER
NR-Aromatase
NR-AhR
NR-AR-LBD
NR-AR
% Positive (toxic)
ESOL MAE (norm)
Train
Val
Epoch
Lipophilicity MAE (norm)
Train
Val
Epoch
Tox21 AUROC
Train
Val
Epoch


Solubility optimization (logS)
logS (mol/L)
Step
Target: -1.62
Start: -3.12
Lipophilicity optimization (logD)
logD
Step
Target: 2.50
Start: 1.67
Lipinski OK
Toxicity minimization (mean prob)
Mean tox probability
Step
Start: 0.027
Combined objective loss
Loss
Step

🧬 GNN Drug Discovery Platform

Graph Neural Networks for Molecular Property Prediction & Inverse Design

Project	Dataset	Task	Model
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Project 2	ESOL · Lipophilicity · Tox21	Multi-property prediction	GAT (3 models)

Enter any molecule as a **SMILES string** and explore its predicted properties.

⚛ HOMO-LUMO Gap 🔬 Inverse Design 💧 Solubility & Lipophilicity •••

Project 2 — Predicts water solubility (logS) and fat affinity (logD).

SMILES

CC(C)Cc1ccc(cc1)C(C)C(=O)O

Examples

CC(=O)Oc1ccccc1C(=O)O

CC(=O)Nc1ccc(O)cc1

CC(C)Cc1ccc(cc1)C(C)C(=O)O

Cn1cnc2c1c(=O)n(c(=O)n2C)C

c1ccc2ccccc2c1 C1=CC=CC=C1 CC(=O)O

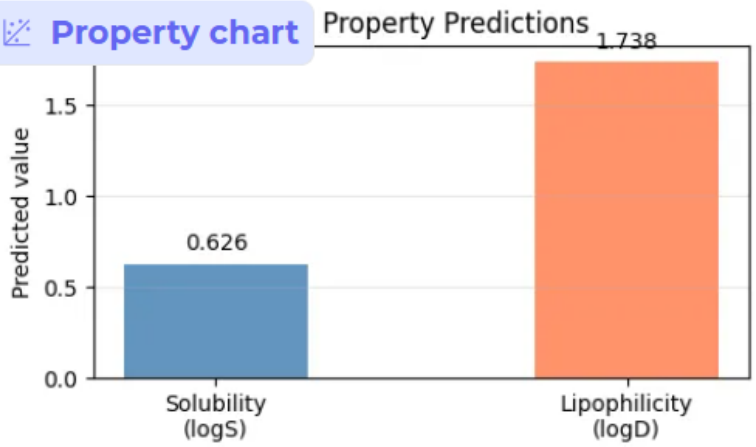
💧 **Predict**

Results

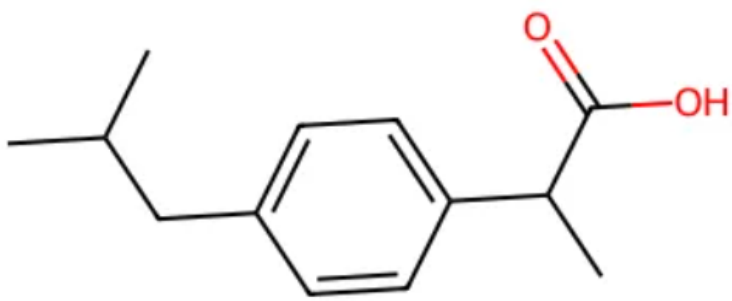
```
═══════════════════════════════════════
Multi-Property Prediction
═══════════════════════════════════════
Solubility (logS)   : 0.6263  🟢 Good
Lipophilicity (logD) : 1.7377  🟢 OK

RDKit descriptors:
  MW   : 206.28 g/mol
  TPSA : 37.30 Å²
  HBD  : 1
  HBA  : 1
═══════════════════════════════════════
```

Property chart



Molecule



HOMO-LUMO Gap | Inverse Design | Solubility & Lipophilicity

Project 1 — Gradient ascent on molecular graph toward a target HOMO-LUMO gap.

Seed molecule (SMILES)

CC(=O)Nc1ccc(O)cc1

Target HOMO-LUMO gap (eV) 4

0.5 — 11

Optimization steps 350

50 — 500

Learning rate 0.04

0.01 — 0.15

Example seed molecules

- CC(=O)Oc1ccccc1C(=O)O
- CC(=O)Nc1ccc(O)cc1
- CC(C)Cc1ccc(cc1)C(C)C(=O)O
- Cn1cnc2c1c(=O)n(c(=O)n2C)C
- c1ccc2ccccc2c1
- C1=CC=CC=C1
- CC(=O)O

Run Inverse Design

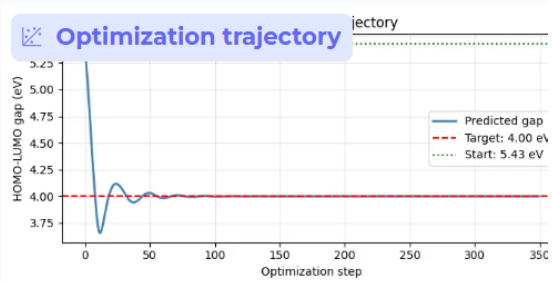
Results

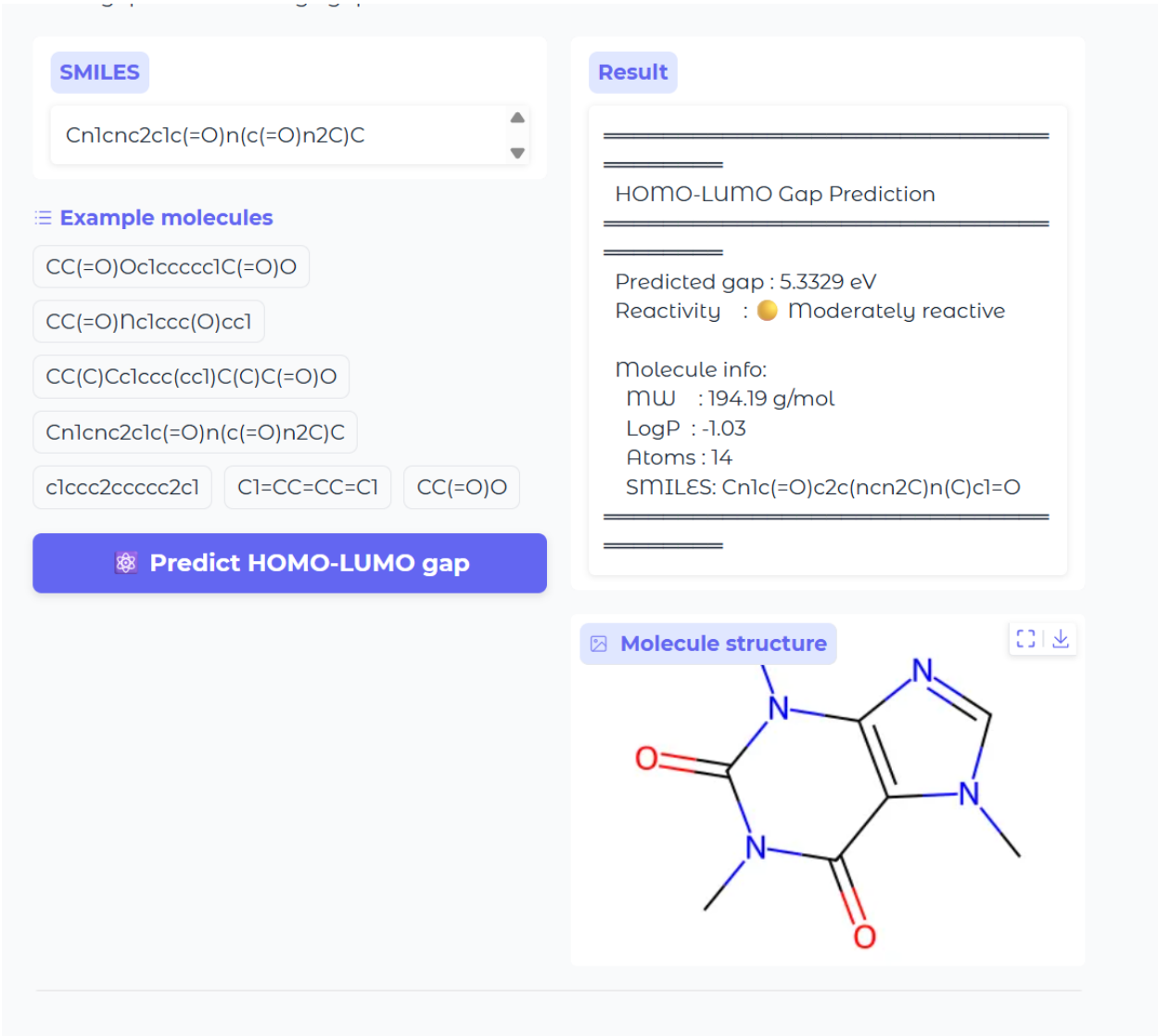
```
Inverse Design Results

Seed SMILES  : CC(=O)Nc1ccc(O)cc1
Initial gap  : 5.4295 eV
Target gap   : 4.0000 eV
Final gap    : 4.0000 eV
Change       : -1.4295 eV
Steps run    : 350
```

Note: Optimized features are continuous. Decoding to a valid SMILES requires additional graph-to-molecule postprocessing.

Optimization trajectory





9. Convergence Analysis

- Training loss decreases steadily over epochs
- Validation loss stabilizes after ~40-60 epochs
- No major overfitting observed
- Inverse design: Gap converges gradually toward target with stable optimization trajectory

10. Error Analysis

Observed Errors:

- Slight overestimation of lipophilicity (logD)

- Perfect inverse outputs (over-optimization)

Reasons:

- Limited dataset size
- Lack of strong chemical constraints
- Continuous optimization space

11. Comparison with Baseline Model

Model	Performance
Linear Regression	Poor
MLP (no graph)	Moderate
GNN (GCN/GAT)	Best - Recommended

GNN outperforms baseline models because it captures the full molecular structure and graph topology.

12. Observations and Inferences

- GNN successfully learns chemical patterns (Conjugation -> lower gap, Polarity -> higher solubility)
- RDKit descriptors improve interpretability
- Multi-property prediction increases real-world usefulness

13. Challenges Faced and Solutions

Challenge	Solution
Invalid SMILES	Filtering using RDKit
Dataset handling	Efficient PyG loaders
Overfitting	Dropout + early stopping
Inverse design instability	Step control + learning rate tuning
CPU limitation	Lightweight model design

14. Future Enhancements

- Add binding affinity prediction
- Use larger datasets (PubChem, ChEMBL)
- Improve inverse design with graph decoding and chemical constraints
- Integrate LLM + GNN (GALLON approach)
- Deploy as a web application

Final Summary (For Viva)

- Uses Graph Neural Networks to understand molecules
- Predicts important chemical properties
- Generates new molecules via inverse design
- Evaluates drug-likeness using ADMET analysis

This project demonstrates a complete AI-driven drug discovery pipeline combining graph learning, chemical informatics, and optimization techniques to predict and design molecules with desired properties.